AUDIO MACHINE LEARNING · DEMO NOTEBOOK

Real-Time Speech Enhancement using PyTorch

Turning **noisy** speech back into **clean** speech with a small CNN that learns a time-frequency mask — data & a baseline model pulled from Hugging Face.

download → add noise → STFT → CNN mask → reconstruct

THE TASK

What is speech enhancement?

Given a recording where speech is buried in noise, produce a cleaner version where the voice stands out.

Noisy speech
voice + background noise

→

Enhancer
what we build

→

Clean speech
noise removed

Used in calls, voice assistants, hearing aids, transcription and meeting tools — anywhere clearer voice helps.

THE APPROACH

Spectral masking

Speech and noise look different in a spectrogram. We train a network to predict, for every time-frequency point, **how much is speech** (a mask from 0 to 1), then multiply it back in.

Noisy wave 16 kHz → **STFT** magnitude + phase → **CNN** predicts mask 0..1 → **mask × mag** keep noisy phase → **ISTFT** clean wave

Why it works: the mask simply turns down the bins dominated by noise and keeps the ones dominated by voice.

THE MODELS

One we train, one we borrow

MODEL	WHAT IT IS	SOURCE
Our CNN	A tiny fully-convolutional mask predictor — just **11k parameters**. We train it on our noise.	built in the notebook (PyTorch)
MetricGAN+	A strong pretrained enhancer, trained on **real-world** noise & tuned for perceptual quality.	`speechbrain/metricgan-plus-voicebank`

Both are **spectral-mask** models, so the comparison is apples-to-apples. The pretrained one is pulled straight from the Hugging Face Hub.

THE DATA

Clean speech from Hugging Face + synthetic noise

Clean speech (the Hub)

73 LibriSpeech clips (~8 min, 16 kHz) decoded straight from a Parquet file — no heavy download.

repo: `hf-internal-testing/librispeech_asr_dummy`

Noise we control

We add Gaussian white noise at a chosen SNR to make `(noisy, clean)` pairs.

- Random SNR per sample while training (augmentation)
- Fixed **0 dB** for the final, fair comparison

61 clips → 242 training segments of 1.5 s; 12 clips held out for evaluation.

THE NOTEBOOK

End-to-end in 11 steps

1–2 · Setup

Install deps, configure STFT & training.

3 · Download

Clean speech from Hugging Face.

4 · Add noise

Mix Gaussian noise at a target SNR.

5 · STFT / ISTFT

Wave ↔ spectrogram (magnitude + phase).

6–7 · Data + CNN

Build pairs & the mask network.

8 · Train

Fit the mask with MSE on CPU.

9 · Reconstruct

ISTFT + SI-SNR + listen.

10 · Compare

MetricGAN+ from the Hub.

11 · Results

Bar chart + real-time factor.

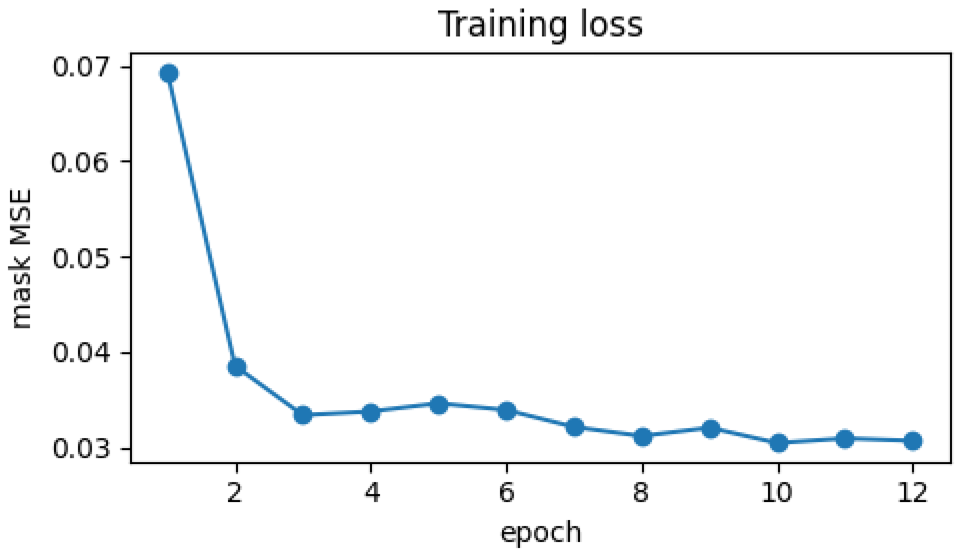
TRAINING

A tiny model, trained on a laptop CPU

11k
parameters
4 conv layers

199s
to train
12 epochs, no GPU

0.015
real-time factor
≈66× faster than real time

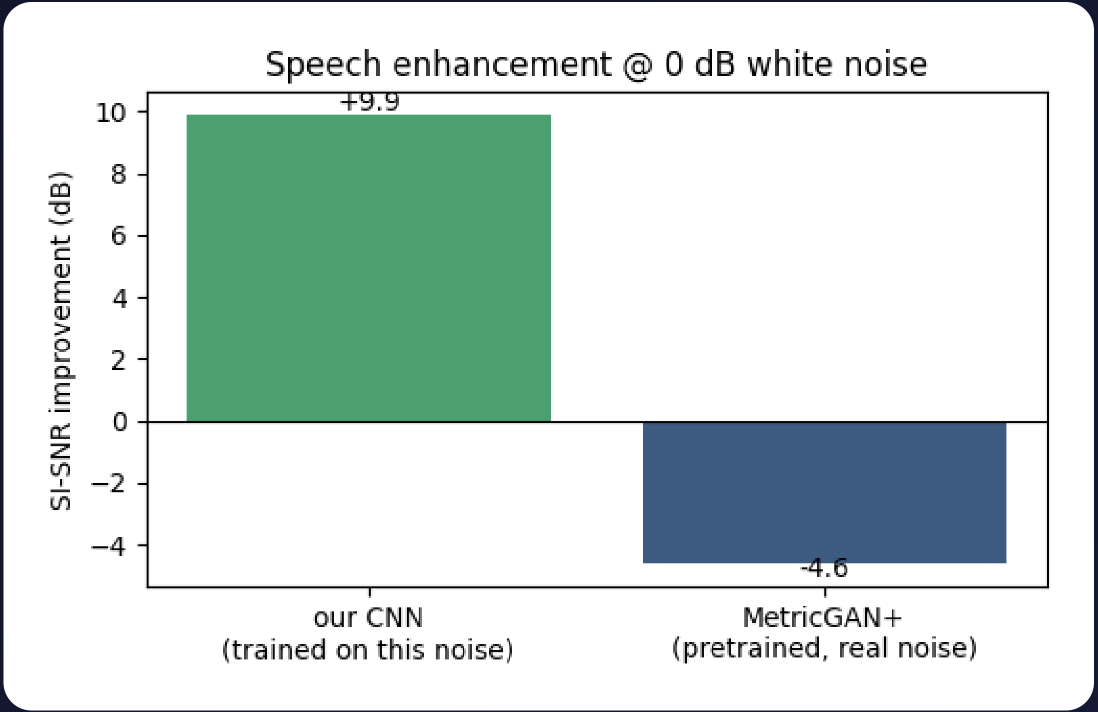


RESULTS

SI-SNR improvement @ 0 dB white noise

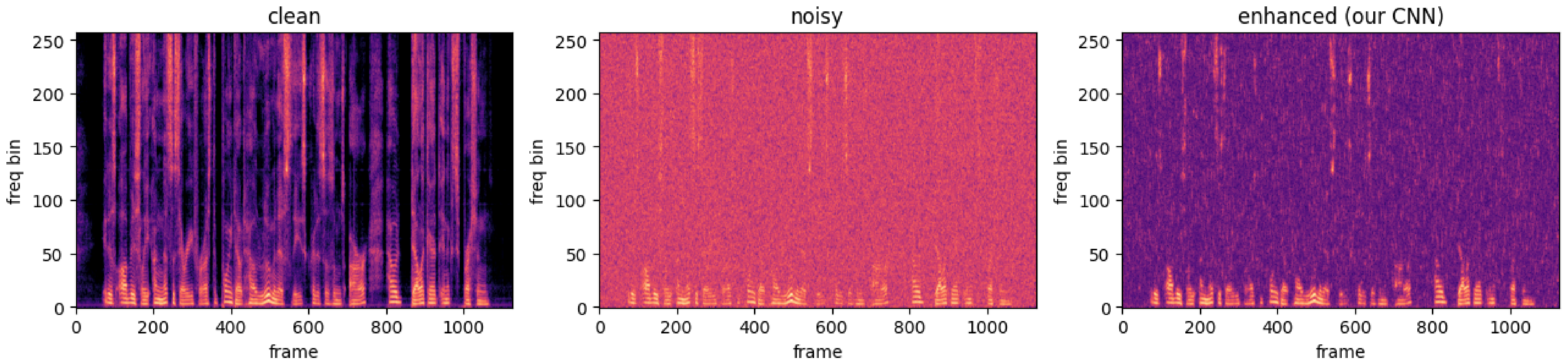
METHOD	SI-SNR IMPROVEMENT
Our CNN	**+9.9 dB**
MetricGAN+ (pretrained)	**−4.6 dB**

Higher is better. Our small CNN, trained on exactly this noise, cleans it up dramatically; the pretrained model — built for different, real-world noise — does not help here.



RESULTS

You can see the noise disappear



Left to right: **clean**, **noisy** (haze everywhere), and the **CNN-enhanced** output — the speech structure is recovered and the background haze is suppressed.

THE KEY INSIGHT

A pretrained model only helps on the noise it knows

Why our CNN wins here

It was trained on the *exact* white noise we test on — so it specializes and removes it well.

Why MetricGAN+ struggles

It learned *real-world* recorded noise and optimizes perceptual quality — white noise is out of its domain, so SI-SNR even drops.

Takeaway: match the model to your noise. A small task-specific model can beat a big general one on its own turf — and on real recordings, MetricGAN+ would be the stronger pick.

TAKEAWAYS

What this demonstrates

- STFT + a simple CNN mask is a complete, understandable speech-enhancement recipe.
- An 11k-parameter model gains **+9.9 dB** SI-SNR on its target noise.
- It runs 66× faster than real time on CPU — genuinely "real-time".
- Data *and* the baseline model come straight from Hugging Face.

NEXT STEPS

How to make it stronger

Real noise

Train & test on recorded noise (ESC-50 / DEMAND) so it generalizes — then MetricGAN+ becomes a fair rival.

More data & context

Larger speech sets; feed several frames or log-mel features for context.

Bigger model

Swap the plain CNN for a U-Net or add recurrent layers.

Better objective

Train directly on an SI-SNR loss instead of mask MSE.

THANK YOU

Questions?

Notebook: `notebooks/speech_enhancement/speech_enhancement.ipynb`

Data & baseline model: Hugging Face Hub · LibriSpeech · SpeechBrain MetricGAN+

Run it top to bottom — trains in ~3 minutes on CPU